

Lab 6

Monday, October 20, 2025 11:25 AM

Lab 6: PWM Controller for a RGB LED

Objective

The goal of this lab is to design, and implement a PWM controller for a RGB LED.

Over UART the HW will receive a sRGB pixel -> R,G,B.

Then it will light the LED with the given pixel color.

The system will feedback to the PC the translated values from sRGB to LED values.

This lab is heavily based on previous labs - do not proceed with this labs without completing the previous labs.

The code must be in SystemVerilog, parametrized and generate based [wisely](#).

Requirements

1. Inputs / Outputs

- Inputs:
 - System clk - 100MHz
 - CPU Reset Push-Button : 1-bit negatively asserted
 - Center/Up/Down/Left/Right Push-buttons: 1 bit
 - RX serial line
- Outputs:
 - 8-Digits 7-Segments: Cathodes & Anodes
 - RGB outputs for two three-colors LEDs: LED16, LED17
 - LED16_B, LED16_G, LED16_R
 - LED17_B, LED17_G, LED17_R
 - LEDS:

[5:4]	T0
[8:7]	T01
[10:9]	T2
[13:12]	T03
- TX serial line

2. UART PHY

- Baud rate should be defined within our package
 - default: 57600 bps

3. UART MAC

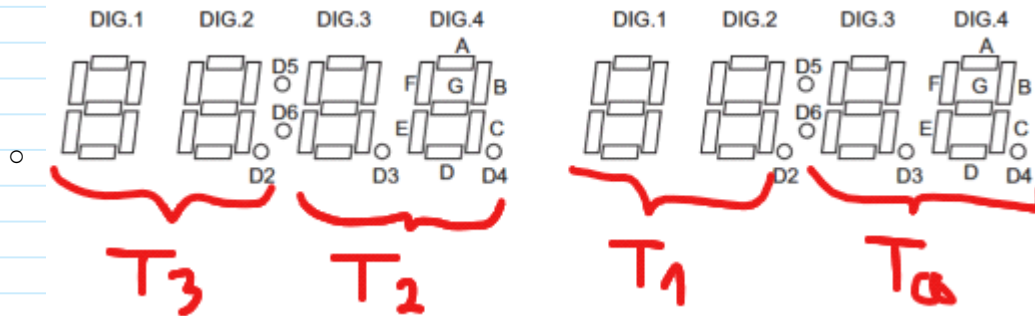
- Two supported messages: Pixel values, Led selector
 - {R_ __, G_ __, B_ __} - Pixel Values [sRGB]
 - {L_ __} - Led Selector ("016" or "017")

4. Push-buttons

- Each push button should be debounced (20ms debouncing window)
- Generic code + Generated instantiation
- The buttons will control the selected argument for change (Led/R/G/B)
 - Up/Down - Will inc. or dec. the value (no wrap around!)
 - Right/Left - Will move the argument selector right/left (no cyclic shifting!)
 - Center push button will apply new values for PWM controller

5. 7-Segment to display values - TX Mode

- Display should not flicker



- T0: Received/Set Pixel's **Blue** Value [Hex]
- T1: Received/Set Pixel's **Green** Value [Hex]
- T2: Received/Set Pixel's **Red** Value [Hex]
- T3: Received/Set LED:
 - "01" -> LED16
 - "10" -> LED17

6. PWM Controller:

- Describe in your report the used PWM slots:
 - 1024, but parametrized to support 512 or 2048,
- Translate sRGB values to LED Linear values (Gemma correction) - Use the provided module
- Apply CIE scaling to avoid LED intensity

7. Testbench:

- Use timescale 1ns / 1ps.
- Use defines or parameters for controlling real HW or Simulation

8. Physical Implementation:

- Set the XDC file with the abovementioned I/Os
- Connect the dut ports appropriately
- Synthesize, PnR, Gen bitstream
- Test on the Nexys board

9. Deliverables:

- Vivado Project
- Verilog code for the design and the testbenches.
- Validated Bitstream
- Simulation waveform screenshots (paste into your report)
- A report with:
 - Blocks diagram
 - FSM design diagrams
 - Lint schematics and clean report
 - Synthesis & Implementation reports are clean
 - Resource expectations & Resources usage report
 - are the reported resources meet expectations?
 - Design process and simulation results

